Faithful by Construction? An Exact Test of LLM Explanations for Constrained Portfolio Optimization Decisions

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Abstract

When a constrained optimizer selects a portfolio, managers, auditors and regulators ask *why*. Large language models (LLMs) are increasingly used to answer in plain language, but a fluent explanation can cite reasons that did not drive the decision. In a linear program, the effect of every constraint on the optimum is known from the solver's certificate and can be confirmed by relaxing the constraint and re-solving. Each claim in an explanation ("this limit drove the allocation and is worth about x basis points") can therefore be checked against ground truth rather than against another model or a human judge. We apply this test to 196 bond-portfolio problems built from public FRED data on US Treasury yields and ICE BofA credit spreads (May to October 2026), scoring explanations written by Claude Sonnet 4.5 and Gemini 2.5 Flash without access to the certificate. Across 149 cases in which a limit had exactly zero effect on the optimum, the models named it as a driver in 89% (Claude) and 77% (Gemini) of cases, while naming binding limits in 98% and 90% of cases. Giving the models the portfolio facts that show which limits are tight left Claude's rate essentially unchanged (86%) and reduced Gemini's to 48%, no better than a fixed template that always names the same two limits (50%). The models' estimates of what each binding limit costs were within $\pm$35% of the true value in 7% of 1,703 claims, with median errors ranging from $0.29\times$ to $8.43\times$ the true value and in the same direction for both vendors, even though the holdings shown to the models determine the true values. A third model, GPT-4o-mini, copied the numbers in the prompt's format example and is reported separately. Explanations generated from the certificate avoid both errors by construction.

Keywords: explainable optimization; faithfulness; large language models; portfolio optimization; shadow prices; linear programming; model risk.

JEL classification: G11, C61, C45, G23, C63.

AI disclosure. The author used Claude (Anthropic) for coding assistance, literature search and editing. All results were checked against the saved model outputs, and the author is responsible for the content. An Anthropic model, Claude Sonnet 4.5, is among the models evaluated.

Data and code. Code, model replies, scored results and the scripts that produce every number, table and figure are available at `https://github.com/Prateek2690/faithful-by-construction`. The ICE BofA spread data are not redistributed. The repository includes a script that downloads all series from FRED, and the problem instances are rebuilt from that download and fixed random seeds.

1 Introduction

Institutional portfolios are increasingly built by optimizers. An objective such as portfolio yield is maximized subject to a mandate: limits on exposure to any single issuer, a ceiling on interest-rate risk, a floor on average credit quality. The solver returns an allocation but not a reason that a portfolio manager, client or examiner can read. A natural shortcut is to ask a large language model (LLM) to explain the allocation. LLMs write fluent and confident explanations, and the question is whether those explanations are *true*.

The interpretability literature separates an explanation that is *plausible* from one that is *faithful* to the process that produced the decision [14, 21], and documents that LLM explanations are often plausible but unfaithful [24, 22]. Testing faithfulness is hard when the decision-maker is a neural model. One must intervene on the cause an explanation names and see whether the decision moves, and the effect of that intervention can only be estimated [3, 25].

Linear programming is a setting where the test can be exact. The solver's optimality certificate states which constraints bind and the marginal value (shadow price) of relaxing each one [5]. A claim of the form "limit i drove the allocation and is worth about x basis points" can therefore be checked by relaxing limit i and re-solving. If the optimum does not move, the limit did not drive the decision and the claim is false, however persuasive the surrounding prose. Exact oracles have recently been used to grade LLM explainers of sequential decision policies [11], and solver-grounded LLM assistants exist for optimization models [18, 7]. We use the linear-programming certificate to measure, claim by claim, how faithfully two widely used commercial LLMs explain constrained portfolio decisions when they reason from the solution alone, without access to the certificate.

Our contributions are as follows.

1. A claim-level test for free-form explanations of linear-programming allocation decisions. It separates attribution to limits whose effect is exactly zero from error in the stated magnitudes. The first classification is exact because the optimal value is concave in each limit (§3).
2. An evaluation on 196 bond-portfolio problems built from public FRED data. We compare explanations read from the certificate, a fixed template, and two LLMs with and without the portfolio facts that reveal which limits are tight (§4 and §5).
3. Evidence that both LLMs cite limits with no effect in most of the cases where that error is possible, name binding and non-binding limits at similar rates, are not reliably corrected by being given the relevant facts, and misstate magnitudes with biases in the same direction for both vendors (§5).

We do not propose a new optimizer or explanation system, and our exactness results apply to linear programs only.

2 Related Work

Faithfulness of explanations. Faithfulness, meaning whether an explanation reflects the reasoning actually used, is distinguished from plausibility [14, 21]. Intervention-based tests judge an explanation unfaithful when changing an input in a way that changes the prediction is not reflected in the explanation [3]. Simulatability asks whether an explanation lets an observer predict the model [8]. Recent work treats faithfulness causally [25], estimates the causal effect of the concepts an LLM's explanation cites [22], and tests the necessity and sufficiency of the factors an LLM names for its own decisions, including in a financial-advisory setting [23]. Chain-of-thought explanations have been shown to misstate the true reason for an answer [24]. In these studies the model explains its own decision and the effect of an intervention is estimated. Here the LLM explains another system's decision and the effect is computed.

Exact oracles for LLM explanations. The work closest in spirit is Gross and Spieker [11], who grade LLM explainers of Markov-decision-process policies against probabilistic model checking, using queries such as which action is optimal in a given state. We apply the same idea to linear-programming allocation, where the oracle is the optimality certificate, and we score the attributions and magnitudes an LLM volunteers in a free-form explanation rather than its answers to targeted queries.

Explainable optimization. Sensitivity analysis and duality give the classical account of why a solution is optimal [5]. Recent work formalizes counterfactual explanations for optimization-based decisions [15, 16], uses linear-relaxation duals to attribute decisions of learned combinatorial policies [17], provides software for post-optimality analysis [10], and argues that explanation should become a first-class concern of operations research [4]. These works construct explanations from the model. We test explanations written without it.

LLMs for optimization and finance. OptiGuide [18] and OptiChat [7] answer practitioners' questions about optimization models by calling solver functions. In OptiChat's ablation, GPT-4.1 answered none of the sensitivity questions correctly once those functions were removed [7]. EOR scores LLM explanations of operations-research what-if changes with an LLM judge and with experts [26]. A separate literature checks whether LLMs *formulate* optimization models correctly, using solver-assisted verification [1], multi-check validation [9], hallucination detection [20] and duality-based falsification tests [19]. A correctly formulated model can still be explained unfaithfully. Financial LLM benchmarks evaluate question answering and reasoning [2]. Our contribution is to measure, for portfolio allocation, what an LLM asserts about a solver's decision when it reasons from the solution alone.

3 Method

3.1 Setup

We consider portfolio programs

$$\begin{aligned} \max_{x} \;\; & y^\top x \\ \text{s.t.} \;\; & d^\top x \le D, \;\; q^\top x \ge Q, \;\; \textstyle\sum_i x_i = 1, \;\; 0 \le x_i \le \kappa, \end{aligned} \tag{1}$$

where x are portfolio weights, y asset yields, d durations, q credit-quality scores, D a duration ceiling, Q a quality floor and κ a single-issuer cap. The three mandate limits (κ, D, Q) are the candidate reasons an explanation can cite.

3.2 Ground truth by re-solving

For each limit c we relax its right-hand side by a step of 0.001 in its own units (0.1 percentage point of cap, 0.001 years of duration, 0.001 of quality), re-solve with HiGHS [12], and express the change in optimal yield in basis points (bp) per stated unit: per $+1$ percentage point of cap, per $+0.1$ years of duration ceiling, and per -0.01 of quality floor. A limit is classified as having zero effect if this change is below 10^{-6} bp.

The optimal value of a linear program is concave in each right-hand side. If relaxing a limit leaves the optimum unchanged, its marginal value is therefore zero for every smaller step as well, and the zero-effect classification is exact up to solver tolerance. For a binding limit, the step effect equals the certificate's shadow price unless the step crosses a breakpoint of the value function, in which case it is smaller. In our problems the two agree within 1% for 87% of binding limits, a difference far smaller than the errors reported in §5.3. We use the re-solve rather than the reported duals because it does not depend on reading the certificate correctly and remains well defined at degenerate vertices.

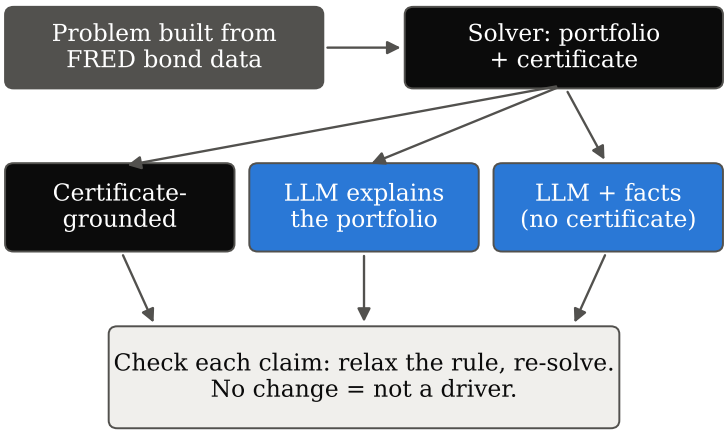


Figure 1: The test. Every claim made by an explainer is checked by relaxing the named limit and re-solving the portfolio problem.

3.3 Explanations as checkable claims

Every explainer returns a set of claims of the form "limit c drove the allocation and relaxing it would add $\hat{\lambda}_c$ bp". We score two things.

- **Zero-effect attribution:** over all (problem, limit) pairs where the limit's effect is zero, the share in which the explainer named the limit as a driver. This is the error most likely to mislead a reader, because it states a reason that provably had no influence. It requires no units or tolerance.
- **Magnitude accuracy:** over claims about limits whose effect is non-zero, the share with $\hat{\lambda}_c$ within $\pm 35\%$ of the true value.

We also report how often binding limits are named, which shows whether an explainer discriminates between limits that matter and limits that do not, and the explanation-level *hard-failure rate*, the share of explanations naming at least one zero-effect limit.

3.4 Explainers

Certificate-grounded. Reads the binding limits and their shadow prices from the certificate and states them. It is correct by construction and serves as a check that the scoring is calibrated, not as a finding.

Template. Always names the issuer cap and the duration ceiling with fixed sizes of 10 and 5 bp. It shows what a constant answer achieves.

LLM, basic. The model sees the constraint limits, every held bond (weight, yield, duration and quality) and the portfolio's yield, but not the certificate. It is asked which limits shaped the allocation and how many basis points relaxing each would add, in the units above, ending with a machine-readable line. The prompt states that "a constraint that is not tight would gain 0 bp and is NOT a driver" and asks the model to include a constraint only if it believes relaxing it would raise yield (Appendix A).

LLM, with facts. As above, plus the realized average duration and average quality next to their limits, and the number of holdings at the cap. These facts are sufficient to tell which limits are tight. The certificate is still withheld.

4 Experimental Setup

Data. We build asset universes from FRED [6, 13]: US Treasury constant-maturity yields at 1–30 years and ICE BofA US corporate option-adjusted spreads for ratings AAA to CCC. Each asset is a representative bond for one (rating, maturity) cell, with yield equal to the Treasury yield plus the rating spread and duration equal to that of a par bond. Quality scores map ratings to $[0, 1]$. This gives 64 assets per date. We use six dates: the month-ends of May to September 2026 and October 14, 2026.

Problems. Each problem samples 25–59 assets from one date's universe and draws the three limits so that, across problems, the duration and quality limits are sometimes tight and sometimes slack. With market spreads all three limits bind in most problems (77 of 96 in our first set), which leaves few cases that test zero-effect attribution. We therefore use two sets: a *natural* set (96 solvable problems, 26 zero-effect cases) and a *slack-focused* set of 100 problems, each with at least one zero-effect limit (123 cases). The issuer cap bound in every problem, so all zero-effect cases concern the duration ceiling (74) or the quality floor (75). Every explainer sees the same problems. In 0 and 3 problems respectively, some limit's measured effect changed by more than 25% between steps of 0.0005 and 0.001, which indicates a breakpoint close to the optimum. These problems are kept.

Models. Claude Sonnet 4.5 and Gemini 2.5 Flash were called through OpenRouter in October 2026 at temperature 0, with one reply per problem and prompt.[1] All replies could be parsed. A third model, GPT-4o-mini, was run in the same way; §5.5 explains why it is reported separately.

5 Results

5.1 A worked example

We chose the following problem to illustrate the error. It is from July 31, 2026 (optimal yield 8.30%). The certificate shows that the issuer cap and the duration ceiling bind: relaxing the cap adds 75.2 bp per percentage point and relaxing the duration ceiling adds 0.76 bp per 0.1 years. The quality floor is not tight, and relaxing it adds nothing. Gemini 2.5 Flash, shown that the portfolio's average quality is 0.535 against a floor of 0.473, answered that *"the credit-quality floor is active, with the portfolio's average quality score just above the 0.47 floor"* and valued it at 8 bp. It valued the cap at 10 bp (true value 75.2) and the duration ceiling at 5 bp (true value 0.76). Each statement reads as reasonable. One names a limit with no effect, and the other two are off by a factor of about seven.

5.2 Attributing the decision to limits that did not matter

Table 1 and Figure 2 give the main result. Across 149 zero-effect cases, Claude Sonnet 4.5 named the limit as a driver in 89% of cases (95% CI 82–93) and Gemini 2.5 Flash in 77% (69–83). The certificate-grounded explainer, as expected by construction, did so in none (upper bound 3%). A template that always names the cap and the duration ceiling scores 50%. This figure reflects the mix of slack limits, since the template names every slack duration ceiling and no slack quality floor.

Discrimination. Claude named binding limits in 98% of cases and zero-effect limits in 89%, and it named all three limits in 87% of its explanations. With the facts, the two rates were 100% and 86%. Gemini with the facts named every binding limit and 48% of zero-effect ones, a partial separation. For comparison, the certificate gives 100% and 0%, and the template 72% and 50%.

Effect of the facts. Showing the models the facts that reveal which limits are tight did not change Claude's zero-effect rate by more than chance would explain (exact McNemar test on paired cases, $p =$

[1] OpenRouter identifiers: `anthropic/claude-sonnet-4.5`, `google/gemini-2.5-flash` and `openai/gpt-4o-mini`.

Table 1: Zero-effect attribution (named / zero-effect cases) in each problem set and pooled with Wilson 95% interval; share of binding limits named; and share of stated magnitudes within ±35% of the true value. *GPT-4o-mini copied the prompt's example values (§5.5).

Explainer	Zero-effect rules named natural	Zero-effect rules named slack-focused	Pooled (95% CI)	Binding rules named	Size within ±35%
Certificate-grounded	0/26	0/123	0% [0, 3]	100%	99.7%
Template: always cap + duration	13/26	61/123	50% [42, 58]	72%	3%
Claude Sonnet 4.5, basic	25/26	107/123	89% [82, 93]	98%	9%
Claude Sonnet 4.5, with facts	23/26	105/123	86% [79, 91]	100%	8%
Gemini 2.5 Flash, basic	17/26	97/123	77% [69, 83]	90%	6%
Gemini 2.5 Flash, with facts	14/26	57/123	48% [40, 56]	100%	7%
GPT-4o-mini*, basic	16/26	72/123	59% [51, 67]	72%	4%
GPT-4o-mini*, with facts	12/26	43/123	37% [30, 45]	95%	9%



Figure 2: Share of zero-effect limits that each explainer named as a driver (pooled over 149 cases; 95% Wilson intervals). Open markers: basic prompt; filled markers: with portfolio facts. *See §5.5.

0.57), but it reduced Gemini's ($p = 1.7 \times 10^{-7}$). With the facts, Gemini was no better than the fixed template (48% against 50%, $p = 0.84$). With the basic prompt, Claude named zero-effect limits more often than Gemini ($p = 0.013$). Because two zero-effect limits in the same problem are not independent, we also report bootstrap intervals that resample whole problems: 83–93 for Claude, 70–83 for Gemini and 40–55 for Gemini with the facts. The natural set, which was not selected for slack limits, gives similar rates (Claude 25 of 26, Gemini 17 of 26), although it contains only 26 cases.

5.3 Misstating how much a limit costs

For limits that did bind, the models' stated magnitudes were within ±35% of the true value in 7% of 1,703 claims (certificate: 99.7%). The errors run in the same direction for both vendors (Figure 3). The issuer cap is understated (median ratio of claimed to true value 0.29×), the duration ceiling is overstated (8.43×), and the quality floor is overstated (2.90×). The result does not depend on the tolerance: 5%, 12% and 17% of claims fall within ±20%, ±50% and a factor of two.

The information needed to compute these values is in the prompt. Held bonds strictly between zero and the cap satisfy $y_j = \mu + \lambda_D d_j - \lambda_Q q_j$, where μ is the budget multiplier and λ_D and λ_Q are the shadow

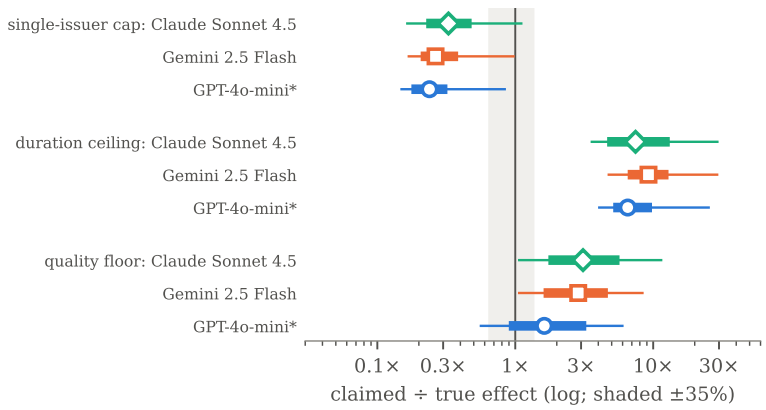


Figure 3: Claimed ÷ true effect of relaxing each limit, per model (median, interquartile range and 10th–90th percentile; log scale). The shaded band is ±35%.

prices of the binding duration and quality limits. These equations determine every binding limit's shadow price in all 439 cases, and the cap's shadow price follows from the bonds held at the cap. Using the rounded values displayed in the prompt, the implied values are within ±10% of the ground truth for 98% of binding limits. The models therefore give plausible numbers rather than computed ones.

5.4 Why the overall error rate understates the problem

On the natural set, the share of explanations naming at least one zero-effect limit was 20% for Claude and 16% for Gemini, against 14% for the template. These rates look moderate only because most limits bind in realistic problems, so naming every limit is usually right. The conditional rates above show that the models do not discriminate: when a limit does not matter, they usually cite it anyway.

5.5 Copying the prompt's example

Our prompt ended with an example of the required output line containing the values 12.0 (cap) and 4.5 (duration). GPT-4o-mini reproduced the 12.0 in 97% of its answers, against 8% for Gemini and 2% for Claude. Because its stated magnitudes largely restate the example, we exclude GPT-4o-mini from the claims above. Its zero-effect rates are reported in full in Table 1: 59% with the basic prompt and 37% with the facts, the lowest of any LLM condition. Even at that rate, more than one in three limits with no effect was cited as a driver.

6 Discussion and Limitations

Implication. An LLM asked to explain a constrained allocation from the solution alone will usually produce a fluent account that cites limits with no effect and misstates the costs of those that matter. Where the explanation becomes a record, such as a committee paper, a client letter or an audit response, this is a governance problem. Explanations generated from the certificate avoid both errors by construction, which is consistent with the design of tool-grounded systems [18, 7]. We did not test whether an LLM that is given the certificate restates it faithfully. That condition is the natural next experiment.

Limitations. (i) Assets are rating–maturity representative bonds calibrated to public data, not individual securities, and the mandate has three limits where real mandates have many more. (ii) The issuer cap bound in every problem, so zero-effect attribution is measured for the duration and quality limits only. (iii) The

exact test relies on linear-programming duality. With integer lot sizes or nonlinear objectives, shadow prices are no longer unique or exact. Magnitudes are measured for a finite step and can understate the local shadow price where a breakpoint is crossed (§3). (iv) We evaluate two models plus one reported separately, with one reply each at temperature 0. Newer models, other sampling settings, reasoning modes or tool use (for example, letting the model call a solver) may behave differently. (v) Our prompt contained an example output line. We report its effect, and a neutral format line is a natural refinement. (vi) The slack-focused set over-represents slack limits by design, and we report the natural set alongside it.

7 Conclusion

For linear-programming portfolio decisions, the drivers of the decision are known exactly, so explanations of those decisions can be checked exactly. Checked this way, explanations written by two commercial LLMs without the solver's certificate named limits with no effect as drivers in most of the cases where that error was possible, while naming binding limits only somewhat more often, and they stated the cost of binding limits within $\pm 35\%$ in only 7% of claims. In settings where an explanation becomes a record, its reasons should come from the certificate rather than from the model's reading of the portfolio.

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A Prompt (with-facts version)

The basic version omits the "Portfolio facts" line. Values are filled in for each problem, and all held bonds are listed (the example below truncates the list). The prompt's "book yield" is the objective $y^\top x$ of (1).

```
A portfolio optimizer maximized book yield subject to these constraints:
  - single-issuer cap: no bond may exceed 4.09% of the portfolio
  - duration ceiling: portfolio average duration must be <= 5.67 years
  - credit-quality floor: portfolio average quality_score must be >= 0.47
  - fully invested: weights sum to 100%

The investable universe has 40 bonds; 25 are held (shown below). Higher quality_score = higher
    credit quality.

Chosen portfolio (book yield = 8.30%):
  bond 1: weight= 4.09% yield=5.86% duration=11.6y quality_score=0.90
  bond 2: weight= 4.09% yield=5.25% duration= 5.7y quality_score=0.80
  bond 3: weight= 4.09% yield=5.44% duration= 4.3y quality_score=0.65
  bond 4: weight= 4.09% yield=5.27% duration= 1.9y quality_score=0.65
  ... (remaining held bonds omitted here)

Portfolio facts: average duration = 5.668y (ceiling 5.668y); average quality_score = 0.535 (floor
     0.473); 23 of 25 held bonds are exactly at the 4.09% cap.

Explain in 3-5 sentences why this portfolio looks the way it does: which
constraints actively shaped it, and for each of those, approximately how many
basis points of book yield could be gained by relaxing it, measured in these
units:
  - issuer_cap: bp per +1 percentage point increase in the single-issuer cap
  - duration_ceiling: bp per +0.1 year increase in the duration ceiling
  - quality_floor: bp per -0.01 decrease in the quality floor
A constraint that is not tight would gain 0 bp and is NOT a driver.

Then, on the LAST line, output ONLY a JSON object of the constraints you judge
to be DRIVERS of the allocation, mapping each to your estimated gain in bp in
the units above. Use exactly these keys where applicable:
"issuer_cap", "duration_ceiling", "quality_floor".
Example final line: {"issuer_cap": 12.0, "duration_ceiling": 4.5}
Include a constraint only if you believe relaxing it would raise book yield.
Output nothing after the JSON.
```

B Reproducing the numbers

All results are computed from the saved model replies, and no API calls are needed. `fetch_fred_universe.py` downloads the FRED series and builds the asset universes (use `--end 2026-10-14` to obtain the data used here; the repository gives its SHA-256 checksum). `run_final_study.py` builds the problems and queries the models (seed 7 for the natural set; seed 11 with `--require_slack` for the slack-focused set). `reproduce.sh` then runs `make_v5_numbers.py`, `make_v5_example.py`, `robustness_v5.py`, `computability_v5.py`, `make_v5_table.py` and `make_v5_figures.py`, which produce every number, the table and the figures. `audit_numbers.py` recomputes the reported figures from the result files and checks them against this PDF.